Yunzhe Daniel **Wang**

AI/ML Research · Agentic AI · Multi-Agent Systems

Los Angeles, CA

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Education

University of Southern California — *Los Angeles, CA*

PhD in Computer Science · Advised by Volkan Ustun and Gale M. Lucas — *Aug 2024 – Expected 2028*

- Research on **self-evolving agents**, **agentic RL**, **LLM-driven social simulation**, **language-agent alignment**, and **agent evaluation**

Columbia University — *New York, NY*

MS in Computer Science, Machine Learning Track · Advised by Hod Lipson — *Aug 2021 – May 2023*

University of Southern California — *Los Angeles, CA*

BS in Computational Neuroscience, BA in Applied Mathematics, Minor in Computer Science — *Aug 2017 – May 2021*

Experience

Amazon Web Services (AWS) – Amazon Quick, Agentic AI Org — *Santa Clara, CA*

Applied Scientist Intern — *Jun 2026 – Present*

- **Mixed User–Agent Co-Evolution** coding agents that learn from many users, privately — NAACL 2027 (submitted)
- **RL Post-Training Environment Synthesis** training worlds and tasks for long-horizon tool-use agents — ACL 2028 (in prep)

University of Southern California — *Los Angeles, CA*

Graduate Research Assistant (PhD Research) — *Aug 2024 – Present*

- **GameplayQA** benchmark for multimodal LLMs that watch many players at once — ACL 2026
- **PersonaEvolve** LLM agents that simulate how crowds act in emergencies — EMNLP 2025
- **FlockWorld** video world model of many agents acting together — NeurIPS 2026 Creative AI
- **DECOY** realistic game simulator for multi-agent planning — WSC 2025, oral
- **X-Ego-CS** team awareness from teammates' first-person views — World Model Open Dataset
- **Steering Room** multi-objective RL LLM agents whose priorities can be adjusted on the fly — ACL 2028 (in prep)
- **GraphAllocBench** benchmark for multi-objective RL agents balancing competing goals — IJCAI-ECAI 2026

Bubble.io — *New York, NY*

Software Engineer II — *Jul 2023 – Jul 2024*

- **AI Page Designer** Bubble's first generative AI product, text to web pages — Public launch 2024
- **Bubble AI** multi-agent LLM system that builds web apps with users — Founding engineer

Columbia University – Creative Machines Lab — *New York, NY*

Research Assistant — *Sep 2021 – May 2023*

- **Robot Lip Sync** humanoid face that learns realistic lip motion — Science Robotics 2026, cover
- **Emo** robot face that smiles back at people — Science Robotics 2024
- **Robot Self-Identification** legged robot that infers its own body from motion — IROS 2024, oral

University of Southern California — *Los Angeles, CA*

Student Researcher — *Feb 2020 – Aug 2021*

- **AI Teammate for Search and Rescue** route planning that supports human teams — AAAI FSS

Open Source

- **ArcForge** community knowledge-graph explorer for the game ARC Raiders — 20+ contributors, 110+ stars
- **X-Ego-CS** multi-player first-person gameplay video dataset on Hugging Face — 66K+ downloads
- **QQ LLM Plugin** LLM reply assistant for the QQ chat app — 70+ stars
- **3D Emoji Assets Generator** turns emojis into game-ready 3D assets — 50+ stars
- **TimeVAE-PyTorch** PyTorch generative model for time series — 40+ stars

Selected Publications

Yunzhe Wang is underlined. † marks a student mentored by Yunzhe.

1. Yunzhe Wang, Runhui Xu†, Kexin Zheng, Tianyi Zhang, Jayavibhav Kogundi, Soham Hans, Volkan Ustun. "**GameplayQA: A Benchmarking Framework for Decision-Dense POV-Synced Multi-Video Understanding of 3D Virtual Agents**." *ACL 2026*. Link
2. Meryem Koksal†, Yunzhe Wang, Volkan Ustun. "**FlockWorld: Toward Population-Scale Multi-Agent Video World Models through Artificial Life Simulation**." *NeurIPS 2026, Creative AI Track*. Link
3. Zhiheng Jiang†, Yunzhe Wang, Ryan Marr†, Ellen Novoseller, Benjamin T. Files, Volkan Ustun. "**GraphAllocBench: A Flexible Benchmark for Preference-Conditioned Multi-Objective Policy Learning**." *MODeM Workshop at IJCAI-ECAI 2026*. Link
4. Yuhang Hu, Jiong Lin, Judah Goldfeder, Philippe Wyder, Yifeng Cao, Steven Tian, Yunzhe Wang, Jingran Wang, Mengmeng Wang, Jie Zeng, Cameron Mehlman, Yingke Wang, Delin Zeng, Boyuan Chen, Hod Lipson. "**Learning Realistic Lip Motions for Humanoid Face Robots**." *Science Robotics 11(110), 2026, Cover Article*. Link
5. Yunzhe Wang, Gale M. Lucas, Burcin Becerik-Gerber, Volkan Ustun. "**Implicit Behavioral Alignment of Language Agents in High-Stakes Crowd Simulations**." *EMNLP 2025, Main*. Link
6. Yunzhe Wang, Volkan Ustun, Chris McGroarty. "**A Data-Driven Discretized CS:GO Simulation Environment to Facilitate Strategic Multi-Agent Planning Research**." *WSC 2025*. Link
7. Yuhang Hu, Yunzhe Wang, Ruibo Liu, Zhou Shen, Hod Lipson. "**Reconfigurable Robot Identification from Motion Data**." *IROS 2024, Oral*. Link
8. Yuhang Hu, Boyuan Chen, Jiong Lin, Yunzhe Wang, Yingke Wang, Cameron Mehlman, Hod Lipson. "**Human-Robot Facial Co-expression**." *Science Robotics 9(88), 2024*. Link

Preprints & Under Review

1. Yunzhe Wang, Erhan Unal, Sheng Zhang, Shabnam Ghadar, Peng Tang. "**Who Remembers What? Multi-Granularity Memory Extraction for Multi-User, Multi-Agent Interaction**." *Submitted to NAACL 2027*.
2. Runhui Xu†, Yunzhe Wang, Volkan Ustun. "**ACG: Ambiguity-Aware Conditional Generation of Team Tactics**." *Submitted to AAMAS 2027*.
3. Tianyi Zhang, Yunzhe Wang, Xiaolin Zhou, Erik Cambria, David Traum, Rui Mao. "**Toward Individualized Cognitive Simulation: Benchmarking and Diagnosing Individual Perspective Alignment**." *Submitted to AAAI 2027*.
4. Yunzhe Wang, Soham Hans, Volkan Ustun. "**X-Ego: Acquiring Team-Level Tactical Situational Awareness via Cross-Egocentric Contrastive Video Representation Learning**." *arXiv:2510.19150, 2025*. Link
5. Yunzhe Wang, Soham Hans, Volkan Ustun. "**The Allocentric Shift: Multi-Ego Contrastive Perceptual Alignment from Coordinated Behavior**." *Under review at ICLR 2027*.
6. Tianyi Zhang, Yunzhe Wang, Volkan Ustun, David Traum. "**Vision-Text Augmentation: Visual Composition as an Optimization Axis for Language Tasks**." *Manuscript in preparation, 2026*.

Mentoring & Service

Mentoring	Meryem Koksal (undergraduate, first author, NeurIPS 2026 Creative AI), Zhiheng Jiang (undergraduate, first author, IJCAI-ECAI 2026 workshop), Ryan Marr (master's, co-author of the same paper), Runhui Xu (master's, second author, ACL 2026)
Reviewing	ICLR 2024–2026, AAAI 2024–2026, EMNLP 2025, NAACL 2027, AAMAS 2025–2027, ICML 2026, NeurIPS 2026
Languages	English (fluent), Chinese (native)